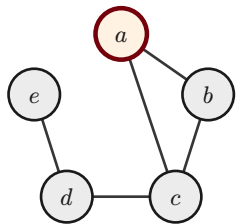
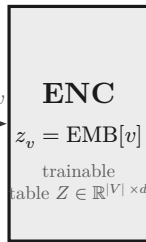


graph $G = (V, E)$



nodes $v \in V$

encoder



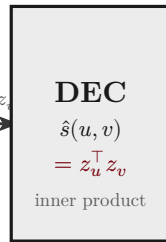
lookup v

Z
 $|V| \times d$

z_a	0.95	0.3
z_b	0.8	0.62
z_c	0.2	0.97
z_d	-0.88	0.4
z_e	-0.55	-0.82

embeddings $z_v \in \mathbb{R}^d$

decoder



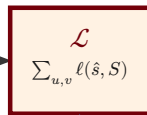
all pairs

$\hat{S} = Z Z^\top$

	a	b	c	d	e
a					
b					
c					
d					
e					

reconstructed sim.

loss



adjacency /
rand-walk



reconstructed proximity $\hat{s}(u, v)$



$\nabla_Z \mathcal{L}$ — update embeddings (SGD)